HEME DEGRADATION AND RELATED DISEASES

Dr. Muhammad Ahsan

MBBS, M.Phil, PFI(St. Petersburg), D.Derm,

Dept. of Biochemistry DMC, DUHS Karachi

OBJECTIVES

- Sites of erythrocytes and hemoglobin degradation.
- Fate of hemoglobin.
- Requirements for Heme degradation.
- Bilirubin formation and its type.
- Excretion of bile pigments through the body.

FATE OF RED BLOOD CELLS

- Life span in blood stream is 120 days
- Senescent(old) RBCs are phagocytosed and/or lysed.
- Normally, lysis occurs extravascularly in the reticuloendothelial system subsequent to RBC phagocytosis
- Lysis can also occur intravascularly (in blood stream).

NORMAL RED CELL DESTRUCTION

- The breakdown of red cells liberates

1- Iron for recirculation via plasma transferrin to marrow erythroblasts

2- protoporphyrin which is broken down to bilirubin.

3- globins which are converted to amino acids.

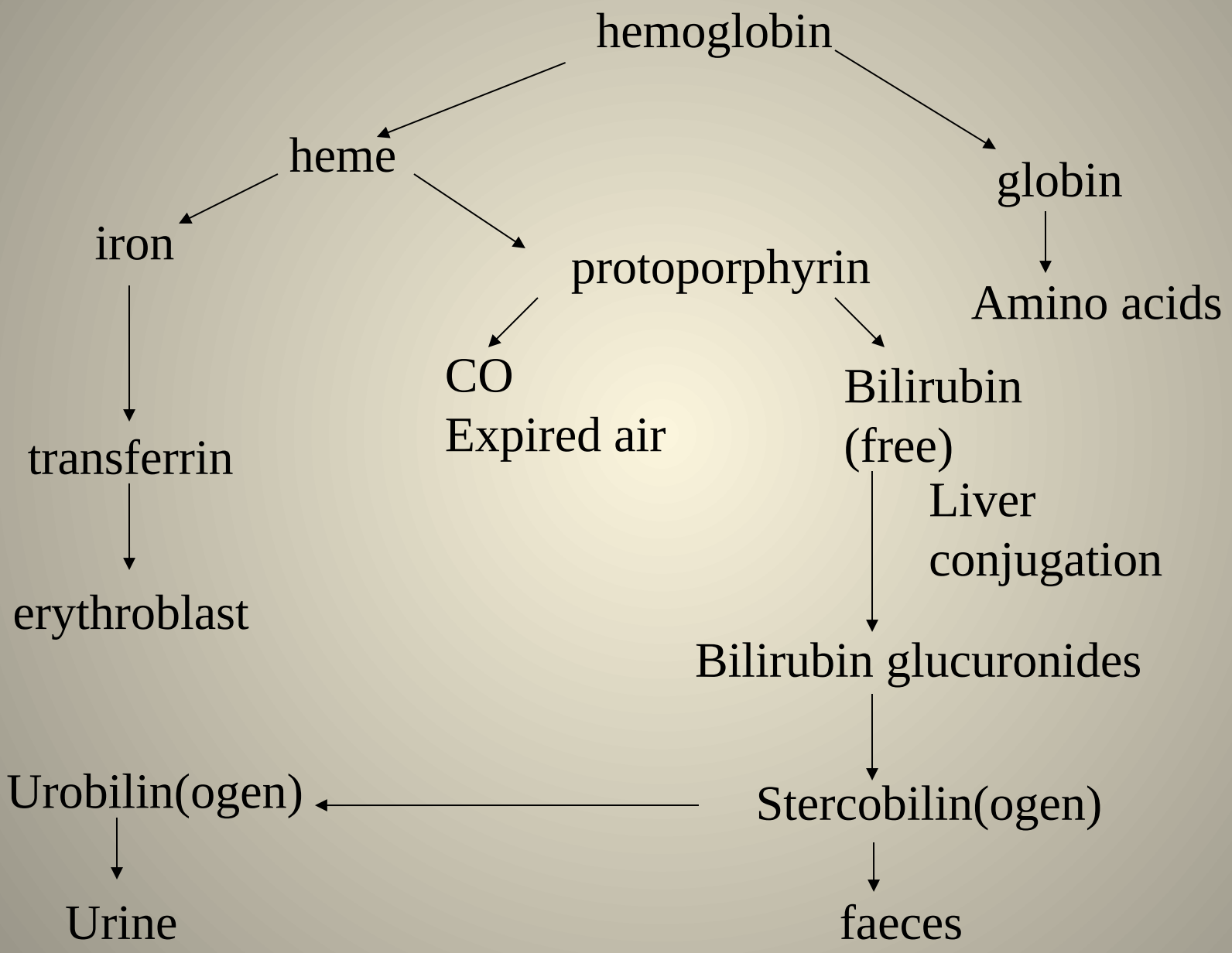
NORMAL RED CELL DESTRUCTION

- The bilirubin circulates to the liver where it is conjugated to glucuronides which are excreted into the gut via bile and converted to stercobilinogen and stercobilin (excreted in faeces).
- Stercobilinogen and stercobilin are partly reabsorbed and excreted in urine as urobilinogen and urobilin.

NORMAL RED CELL DESTRUCTION

- A small fraction of protoporphyrin is converted to carbon monoxide (CO) and excreted via the lungs.
- Globin chains are broken down to amino acids which are reutilized for general protein synthesis in the body.

NORMAL RED CELL BREAKDOWN



Hemoglobin degradation

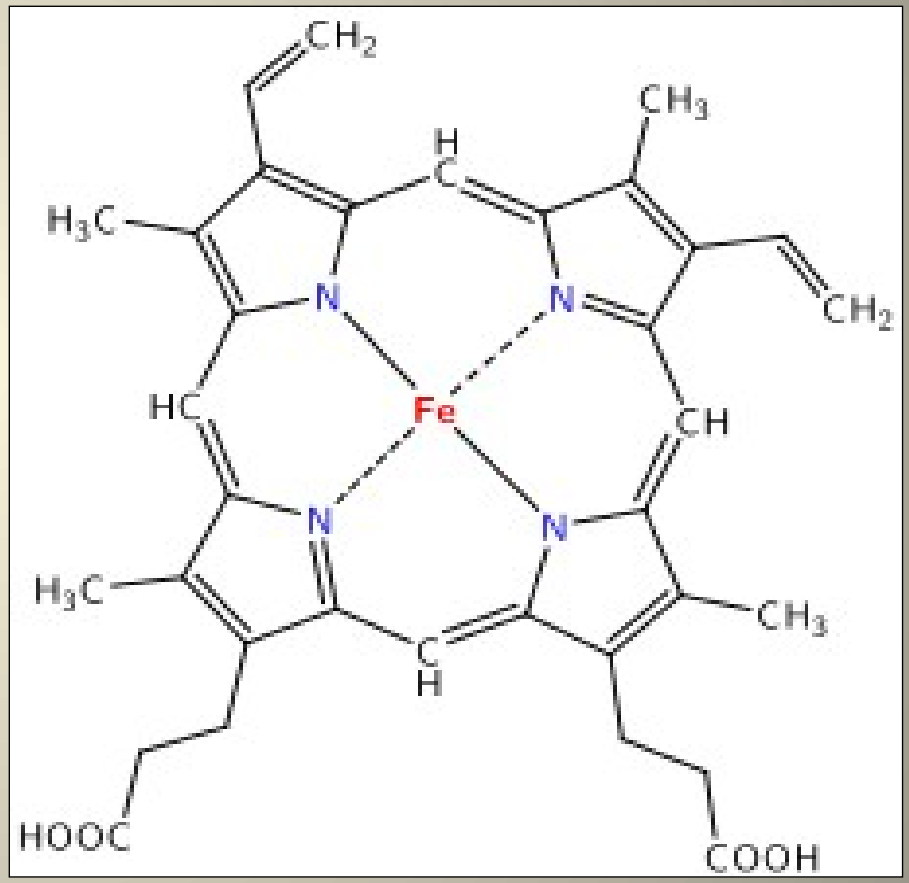
- In the human body approx. 100 – 200 million ery are broken down every hour. Degradation of Hb begins in ER of **reticuloendothelial cells (RES)** of the **liver, spleen, bone marrow** and **skin**.

- Hb is degraded to:
  - globins → AAs → metabolism
  - heme → bilirubin
  - Fe^{2+} → transported with transferrin and used in the next heme biosynthesis

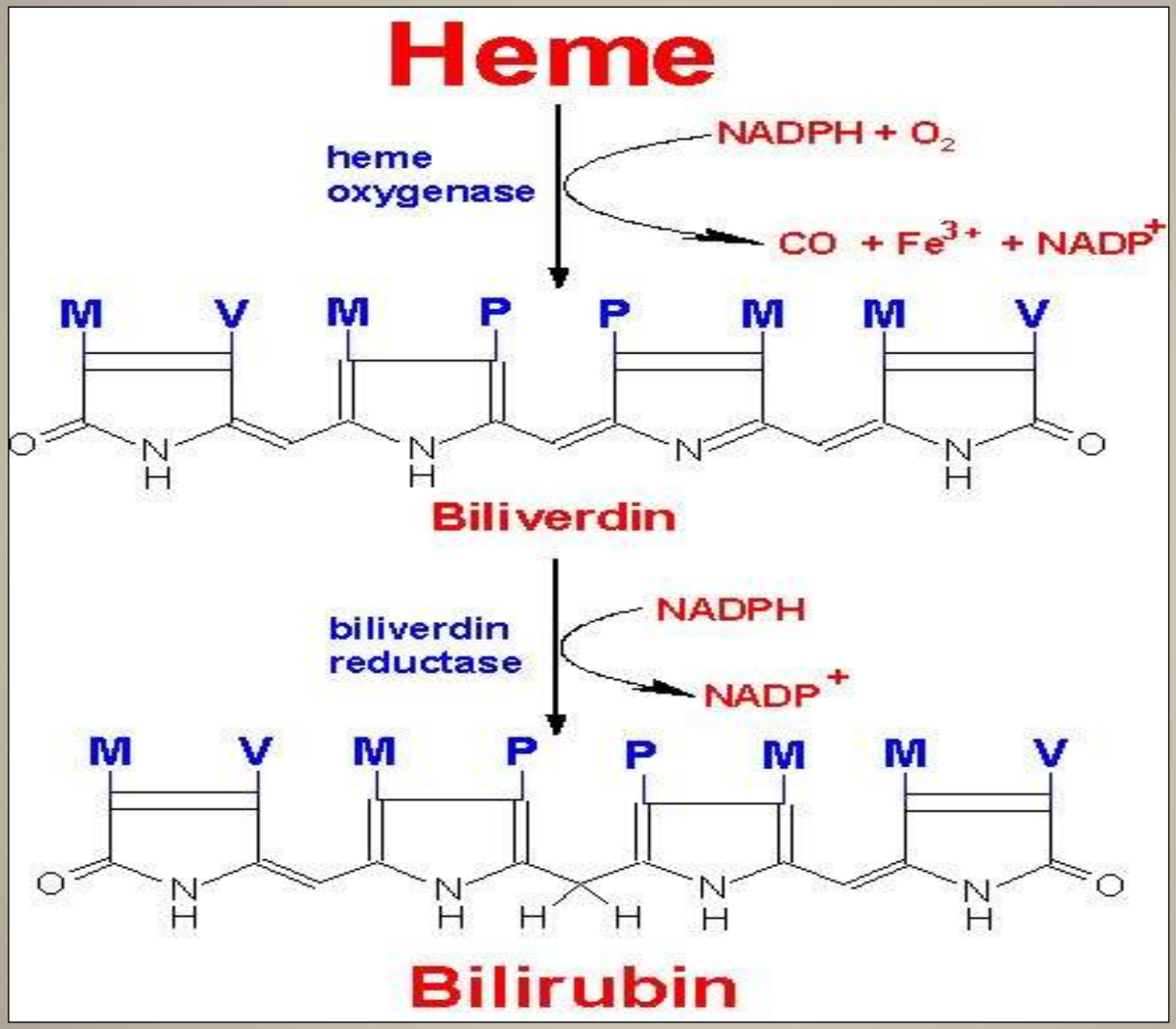
- Not only Hb but other hemoproteins also contain heme groups which are degraded by the same pathway.

Compounds contain heme group

- **Hemoproteins**
- Hemoglobin (Hb)
- Myoglobin (Mb)
- Cytochromes
- Catalases
- Peroxidases



Convertion of heme to bilirubin



green

red-orange

Futher fate of bilirubin

•Bilirubin (Bil) is released from RES into the blood. BUT! **Bil is only poorly soluble in plasma**, and therefore during transport it is bound to **albumin („nonconjugated Bil“).**

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LIVER

•In the hepatocytes, **Bil is conjugated** by 2 molecules of **glucuronic acid → bilirubin diglucuronide (soluble in water, „conjugated Bil“)**

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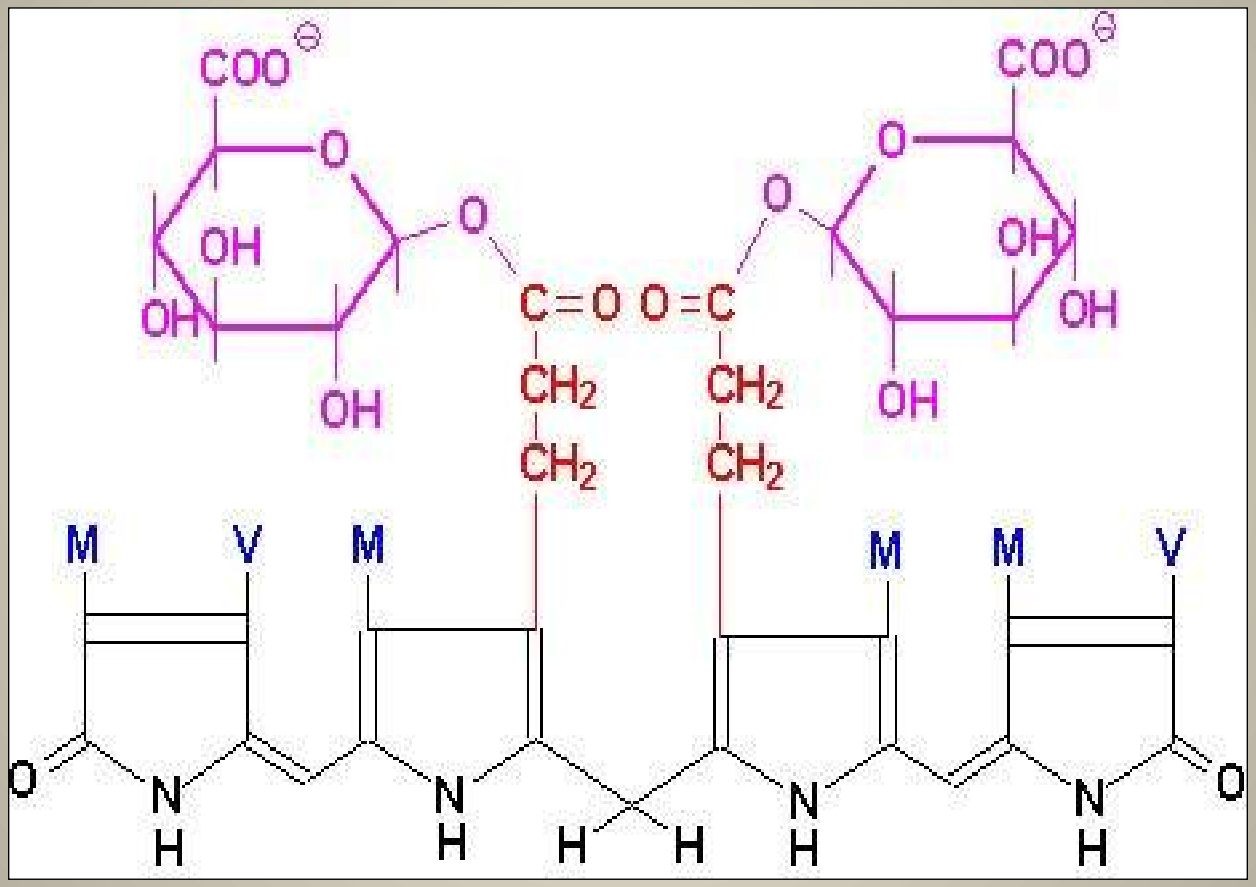
BILE

↓

INTESTINE

•Bil is reduced to **urobilinogen** and **stercobilinogen**

bilirubin-diglucuronide = conjugated bilirubin
is soluble in water → **„direct bilirubin“**



In intestine:

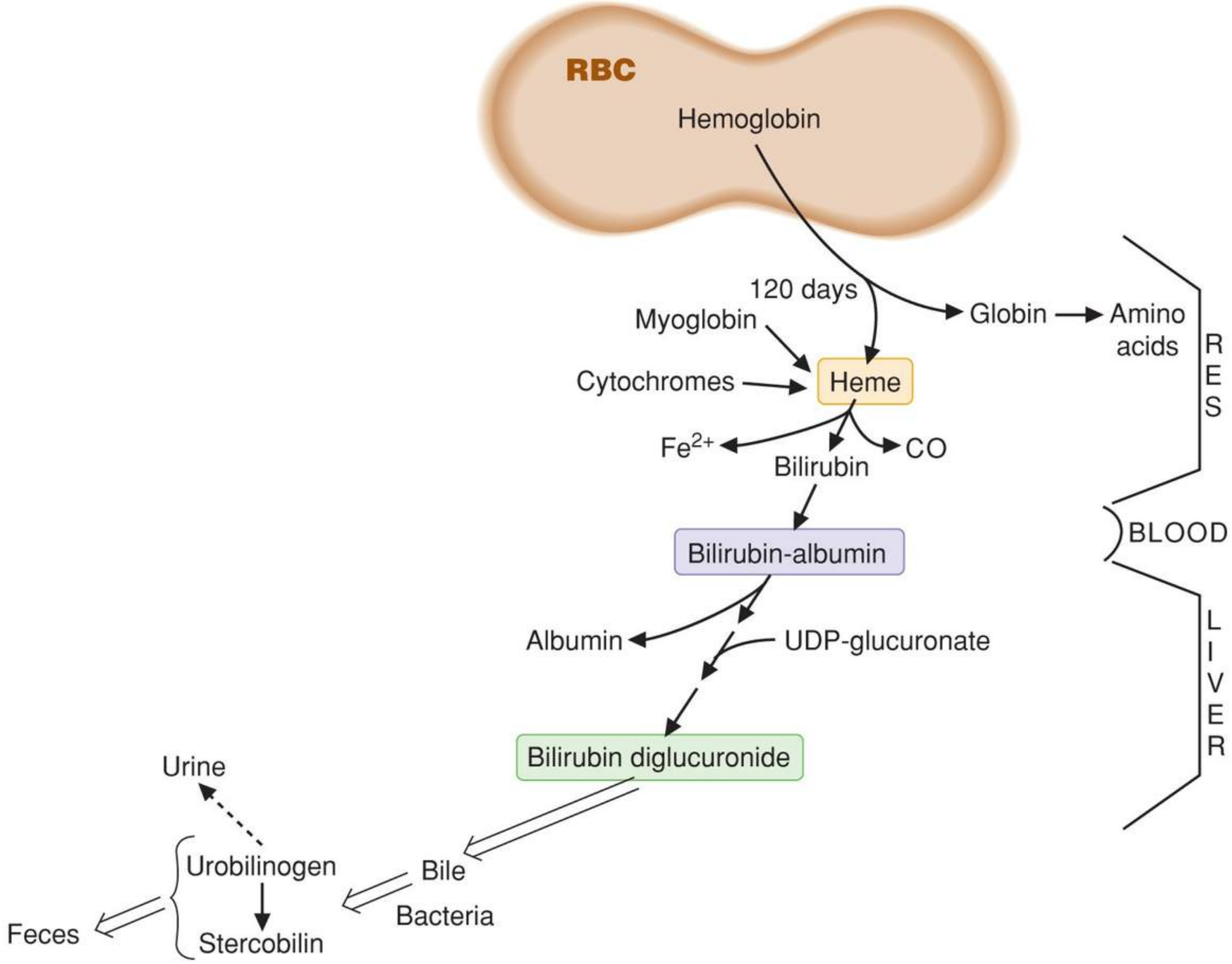
- Urobilinogen and stercobilinogen can be:

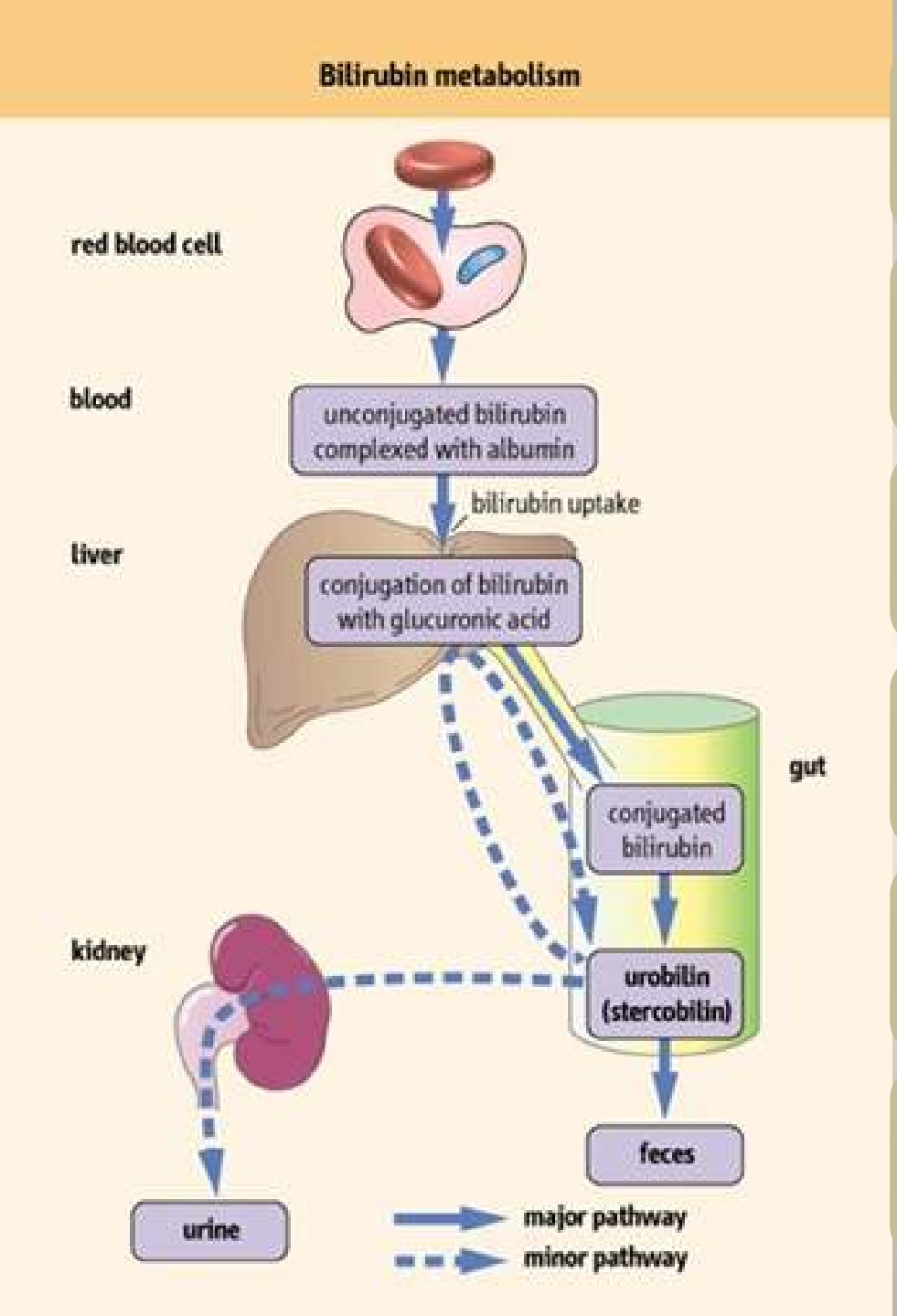
a) reabsorbed and returned to the liver

- **(= enterohepatic circulation)**

b) oxidized (in the presence of O_2) to pigments **urobilin** (orange) and **stercobilin** (yellow) → they are excreted in the stool

- Urobilinogen also appears in the urine.


RBC
Hemoglobin
120 days
Myoglobin
Cytochromes
Heme
Globin
Amino acids
RES
Fe2+
CO
Bilirubin
BLOOD
Bilirubin-albumin
Albumin
UDP-glucuronate
LIVER
Bilirubin diglucuronide
Urine
Urobilinogen
Bile
Bacteria
Feces
Stercobilin



Heme degraded by heme oxygenase in the reticuloendotherial system to bilirubin.

Bilirubin (Hydrophobic) binds to albumin in plasma and transported to liver.

Ligandin serves as the intracellular transporter and carries bilirubin to the endoplasmic reticulum.

Bilirubin is conjugated in the liver to bilirubin diglucoronide (UDP Gluronyl transferase) and excreted in urine.

Bilirubin diglucoronide passes in small intestine where bacterial enzymes produce urobilinogen.

Urobilinogen absorbed and passed to liver where it is re excreted and some amount excreted in urine as urobilin.

Jaundice

Yellow color of skin, nail beds & sclera caused by deposition of bilirubin secondary to increased bilirubin levels in blood (hyperbilirubinemia)

JAUNDICE IS NOT A DISEASE
HOWEVER,
IT IS A SIGN OF
AN UNDERLYING DISEASE



Figure 21.10
Jaundiced patient, with the sclerae of his eyes appearing yellow.

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JAUNDICE:

- Yellow discolouration of skin, sclera and mucous membrane due to abnormal accumulation of bilirubin >3mg/dl.
- TYPES:
  - Pre – hepatic jaundice
  - Intra – hepatic jaundice
  - Post – hepatic jaundice.

PRE – HEPATIC JAUNDICE:

- Also known as Haemolytic jaundice.
- It results from the excess production of unconjugated bilirubin following heamolysis.
- Heamolysis occurs due to:
  - Malaria.
  - Erythroblastosis foetalis
  - Sickle cell anemia
  - Toxins
  - Drugs.

Hepatocellular Jaundice

First

Liver damage (by hepatitis) causes low conjugation efficiency leading to increased unconjugated (indirect) bilirubin in blood

Second

Conjugated bilirubin is not efficiently secreted into bile. Instead, diffuses to blood increasing conjugated (direct) bilirubin in blood

Blood

Increased BOTH unconjugated (indirect) & conjugated (direct) bilirubin

ALT & AST levels are markedly elevated

Urine:

Bilirubin is present in urine

So, urine color is yellowish brown

Stool

Pale (low stercobilin)

HEPATIC JAUNDICE:

- Causes: Primary biliary cirhosis, Alcohal, Drugs, Viral or autoimmune hepatitis, Bacterial infection.

POST HEPATIC JAUNDICE:

- It is also known as Cholestatic jaundice.
- Post hepatic jaundice caused by an obstruction of the biliary tree.
- Increase level of conjugated bilirubin.
- Causes:
  - Stone.
  - Stricture
  - Carcinoma.

Differential diagnosis of jaundice

	Prehepatic	Intrahepatic	Posthepatic
conjugated bilirubin	absent	↑	↑
AST or ALT	normal	↑	normal
ALP	normal	normal	↑
urine bilirubin	absent	present	present
urine urobilinogen	present	present	absent

Neonatal Jaundice

- Common, particularly in premature infants
- Transient (resolves in the first 10 days)
- Due to immaturity of the enzymes involved in bilirubin conjugation
- High levels of unconjugated bilirubin are toxic to the newborn – due to its hydrophobicity it can cross the blood-brain barrier and cause a type of mental retardation known as kernicterus
- If bilirubin levels are judged to be too high, then phototherapy with UV light is used to convert it to a water soluble, non-toxic form

Neonatal Jaundice

- If necessary, exchange blood transfusion is used to remove excess bilirubin
- Phenobarbital is oftentimes administered to Mom prior to an induced labor of a premature infant – crosses the placenta and induces the synthesis of UDP glucuronyl transferase
- Jaundice within the first 24 hrs of life or which takes longer then 10 days to resolve is usually pathological and needs to be further investigated.

Congenital hyperbilirubinemia

Bilirubin is elevated in blood due to inherited defects in the bilirubin metabolic pathway

Crigler-Najjar syndrome

Low activity of glucoronyltransferase (conjugating enzyme)

Rare Inherited l disease

Severe hyperbilirubinemia in neonates (unconjugated bilirubin)

Complicated by kernicterus & early death

Dubin-Johnson syndrome

Defect in transfer of conjugated bilirubin into the biliary canalculi

Conjugated hyperbilirubinemia.

Gilbert`s syndrome

Decreased production (expression) of glucoronyltransferase

Rare autosomal dominant trait

More common men

Occurs in 2-3 % of men

Usually asymptomatic hyperbilirubinemia

Liver function tests are normal

